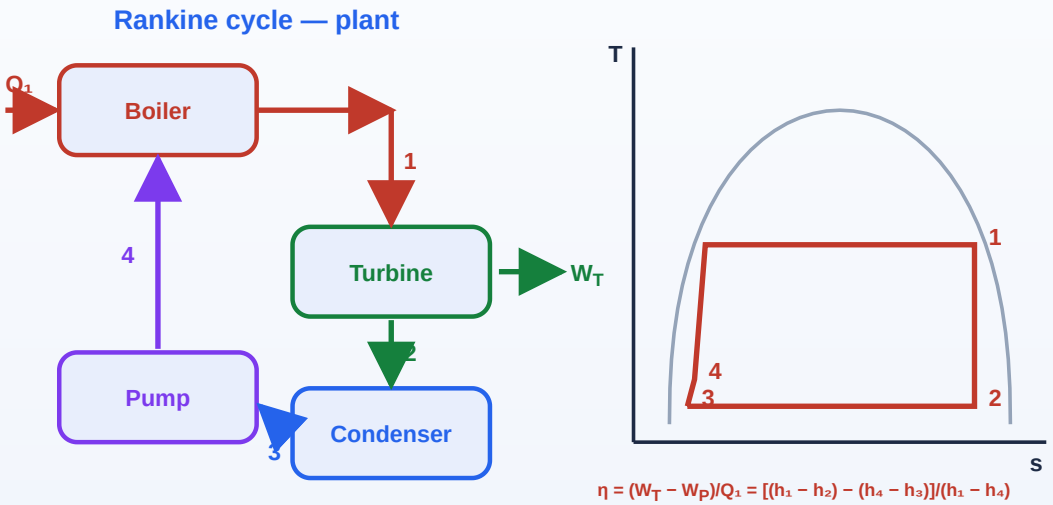
UPSSSC ALERT

UPSSSC JE

MECHANICAL
STEAM & POWER PLANT

Complete English Edition

Unit 6 · 5 Marks · 2026



UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **Steam Engineering and Power Plant Engineering**. Syllabus: Rankine cycle efficiency (with and without pump work), boilers, classification, specifications, mountings, accessories, nozzles and steam turbines.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (R. K. Rajput, P. K. Nag, Domkundwar).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Rankine Cycle

Rankine Cycle – Basic Concept

The ideal cycle of a steam power plant is the **Rankine cycle**. Water is pumped into the boiler, converted to steam, expanded in the turbine to produce work, then condensed in the condenser back to water, which returns to the pump. Most of India's electricity comes from thermal plants based on this cycle.

Understand in Detail — Rankine Cycle

1.1 Steam power plant — the big picture

Most of our country's electricity is produced in coal-fired **thermal power plants**. The idea is simple: burn coal to turn water into high-pressure steam, use the steam to spin a turbine, and the turbine spins a generator. The steam leaving the turbine is turned back into water and sent back to the boiler — and the cycle continues. This ideal cycle is called the **Rankine cycle**.

Four main components: (1) **boiler** — adds heat at constant pressure to turn water into steam (usually superheated); (2) **turbine** — steam expands and produces work; (3) **condenser** — condenses the exhaust steam with cooling water; (4) **feed pump** — pumps the condensate up to boiler pressure. The difference from the Carnot cycle is that the steam is **completely** condensed into water in the condenser, so that only water (not wet steam) has to be pumped — water has a very small volume, so pump work is negligible.

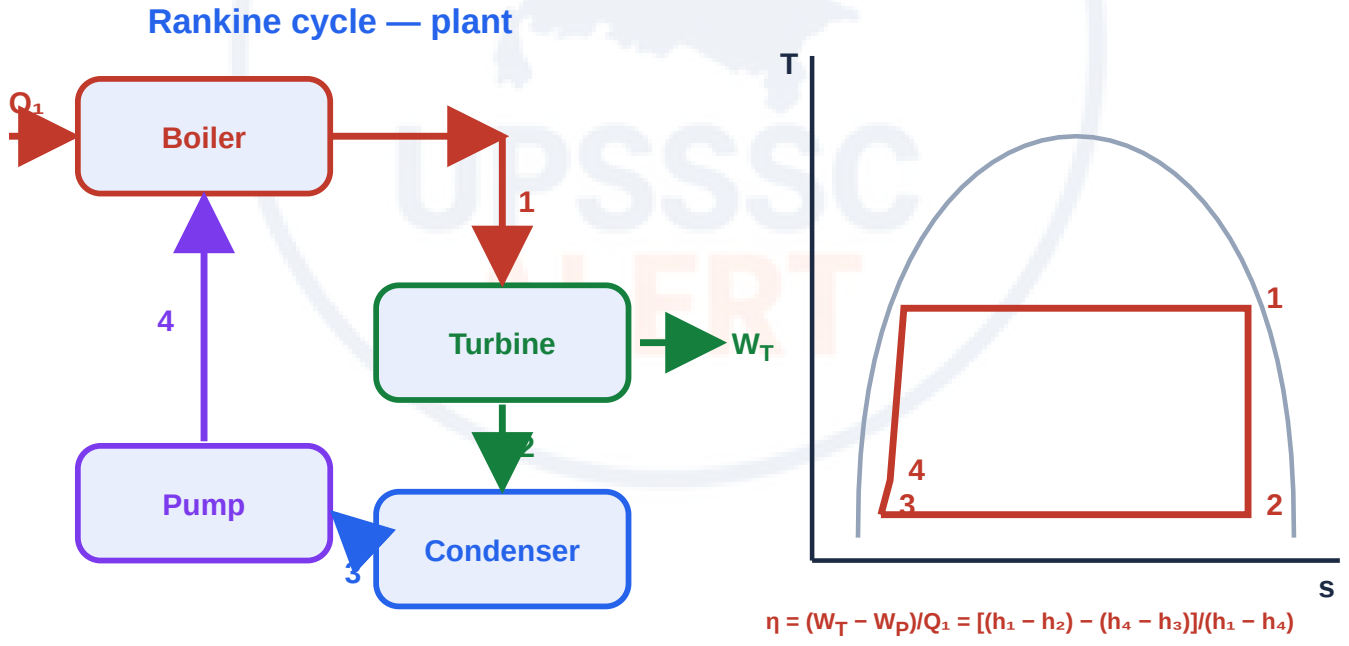


Fig 6.1: Rankine cycle — plant layout and T–s diagram

1.2 The four processes of the Rankine cycle and its efficiency

1 → 2 (turbine): high-pressure steam expands **isentropically** to condenser pressure and produces work: $W_T = h_1 - h_2$. **2 → 3 (condenser):** heat rejection at constant pressure; steam becomes saturated water: $Q_{out} = h_2 - h_3$. **3 → 4 (pump):** isentropic pumping: $W_P = h_4 - h_3 = v_f(p_1 - p_2)$. **4 → 1 (boiler):** heat addition at constant pressure: $Q_{in} = h_1 - h_4$.

Derivation: Rankine efficiency

1. Net work = $W_T - W_P = (h_1 - h_2) - (h_4 - h_3)$
2. $\boldsymbol{\eta_R = [(h_1 - h_2) - (h_4 - h_3)] / (h_1 - h_4)}$
3. Neglecting pump work ($h_4 \approx h_3$): $\boldsymbol{\eta_R \approx (h_1 - h_2)/(h_1 - h_3)}$ — this is mostly used in exams
4. **Work ratio** = net work ÷ turbine work — in Rankine it is nearly 1 (pump work very small), which is a big advantage
5. **Specific steam consumption (SSC)** = $3600/W_{net}$ kg/kWh — steam needed per kWh

Why is Rankine efficiency less than Carnot? In the boiler the water first heats from a low temperature (T_4) — so part of the heat is added at a low average temperature. Carnot receives all heat at the highest temperature. **The higher the mean temperature of heat addition and the lower the temperature of heat rejection, the higher the efficiency** — this is the core principle behind all methods of improving efficiency.

Example: In a Rankine cycle $h_1 = 3200$, $h_2 = 2200$, $h_3 = 190$ kJ/kg; pump work negligible. Find the efficiency and SSC.

Solution:

1. $W_{net} = 3200 - 2200 = 1000$ kJ/kg ; $Q_{in} = 3200 - 190 = 3010$ kJ/kg
2. $\eta = 1000/3010 =$ **33.2%**
3. SSC = 3600/1000 = **3.6 kg/kWh**

1.3 Ways to improve Rankine efficiency — why each one works

(1) Increasing boiler pressure: at higher pressure water boils at a higher temperature → the mean temperature of heat addition rises → efficiency increases. **But** at a fixed maximum temperature, raising pressure makes the steam at turbine exhaust **wetter** — water droplets erode the blades. The dryness fraction at exhaust should not be below 0.88.

(2) Superheating: heating the saturated steam further → mean temperature rises and the exhaust steam stays **drier** — both benefits. Limit: metal endurance (about 600°C).

(3) Lowering condenser pressure: increasing the vacuum lowers the heat rejection temperature → efficiency rises. Limit: cooling water temperature (condensing temperature cannot be lower); a higher vacuum also makes exhaust steam wetter and increases air leakage.

(4) Reheating: steam is partly expanded in the high-pressure turbine, sent back to the boiler's **reheater** to be heated again, then fully expanded in the low-pressure turbine. **Main purpose — reducing exhaust**

moisture; the efficiency gain is small (about 4–5%) but net work increases more. It makes higher boiler pressures usable.

(5) Regenerative feed heating: a little steam is extracted (bled) from the middle of the turbine to heat the feed water in **feed-water heaters**. Now the water reaches the boiler already warm → heat added at low temperature is eliminated → mean temperature rises → efficiency increases. Heaters are **open** (steam and water mix directly, e.g. deaerator) or **closed** (through tubes, no mixing). With infinite heaters Rankine efficiency can approach Carnot.

> **Note:** Summary: **reheating → mainly dryness**; **regeneration → mainly efficiency**. Plants use both together.

1. Key Points

- **Four processes:** 1–2 isentropic expansion in turbine; 2–3 constant-pressure heat rejection in condenser; 3–4 isentropic compression in pump; 4–1 constant-pressure heat addition in boiler.
- **Turbine work:** $W_T = h_1 - h_2$; **pump work:** $W_P = h_4 - h_3 = v_f(p_1 - p_2)$.
- **Efficiency (with pump work):** $\eta = [(h_1 - h_2) - (h_4 - h_3)]/(h_1 - h_4)$.
- **Efficiency (neglecting pump work):** $\eta = (h_1 - h_2)/(h_1 - h_{f2})$ — pump work is very small, so it is often neglected.
- **Work ratio:** net work ÷ turbine work; close to 1 in Rankine (low pump work) — better than Carnot.
- **Specific steam consumption (SSC):** $3600/W_{net}$ kg/kWh.
- **Ways to improve efficiency:** raise boiler pressure, superheat, lower condenser pressure, reheat and regeneration (feed water heaters).
- **Rankine vs Carnot:** compressing wet steam in a Carnot cycle is impractical; Rankine pumps liquid after complete condensation, so it is practical, though less efficient than Carnot.
- **Modified Rankine cycle:** in steam engines expansion is left incomplete (at release), avoiding an excessively long stroke.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Turbine work	$W_T = h_1 - h_2$
Pump work	$W_P = v_f(p_1 - p_2) = h_4 - h_3$
Net work	$W_{net} = W_T - W_P$
Heat supplied	$Q_1 = h_1 - h_4$
Efficiency (with pump)	$\eta = [(h_1 - h_2) - (h_4 - h_3)]/(h_1 - h_4)$
Efficiency (pump neglected)	$\eta = (h_1 - h_2)/(h_1 - h_{f2})$
Work ratio	W_{net}/W_T
Specific steam consumption	$SSC = 3600/W_{net}$ (kg/kWh)
Turbine exit dryness	$s_1 = s_{f2} + x_2\, s_{fg2}$

3. Detailed Explanation

- **Effect of lowering condenser pressure:** mean temperature of heat rejection falls and efficiency rises; but exit steam becomes wetter, causing blade erosion.
- **Superheat and reheat:** superheating raises the mean temperature of heat addition and improves exit dryness; in reheating, partly expanded steam is heated again — mainly to improve dryness.
- **Regenerative cycle:** some steam is bled from the turbine to heat feed water; this raises the mean temperature of heat addition and the efficiency.
- **Turbine exit dryness:** for isentropic expansion $s_1 = s_2$, so $x_2 = (s_1 - s_{f2})/s_{fg2}$; in practice $x_2 \geq 0.88$ is kept.
- **Why pump work is small:** the specific volume of liquid is very small (≈ 0.001 m^3/kg), so $v \cdot \Delta p$ is small.

4. Comparison Table — Rankine vs Carnot Cycle

Basis	Rankine	Carnot
Compression	Pumping of liquid	Compressing wet steam
Practicality	Practical	Impractical
Work ratio	High (≈ 1)	Low
Efficiency (same limits)	Lower	Higher

5. Solved Numerical Problems

Example 1. Rankine cycle: $h_1 = 3200$, $h_2 = 2200$, $h_{f2} = h_3 = 200$ kJ/kg, pump work neglected. Find the efficiency.

Solution:

1. $W_T = 3200 - 2200 = 1000$ kJ/kg; $Q_1 = 3200 - 200 = 3000$ kJ/kg
2. $\eta = 1000/3000 =$ **33.3%**

Example 2. In the above, boiler pressure is 100 bar and condenser pressure 0.1 bar ($v_f = 0.001$ m³/kg). Find efficiency with pump work.

Solution:

1. $W_P = 0.001 \times (10000 - 10) = 9.99 \approx 10$ kJ/kg
2. $h_4 = 210$ kJ/kg; $Q_1 = 3200 - 210 = 2990$ kJ/kg
3. $\eta = (1000 - 10)/2990 =$ **33.1%**

Example 3. Find the specific steam consumption for the above.

Solution:

1. SSC = 3600/990 = **3.64 kg/kWh**

Example 4. Find the work ratio for the above.

Solution:

1. Work ratio = 990/1000 = **0.99**

Example 5. At turbine exit $s_1 = 6.5$, $s_{f2} = 0.65$, $s_{fg2} = 7.5$ kJ/kg·K. Find the dryness.

Solution:

1. $x_2 = (6.5 - 0.65)/7.5 =$ **0.78**

6. Frequently Asked Facts

- Rankine – the ideal cycle of steam power plants.
- Pump work = $v_f \Delta p$, very small; hence work ratio ≈ 1.
- To improve efficiency: high boiler pressure, superheat, low condenser pressure, reheat, regeneration.
- Main purpose of reheating — increase dryness at turbine exit.

7. Note (Confusing Facts)

- Rankine efficiency is lower than Carnot's for the same limits, yet Rankine is practical.
- Lowering condenser pressure raises efficiency but reduces dryness.
- Pump work is $h_4 - h_3$, not $h_3 - h_4$.

Remember: One line: $\eta = W_{net}/Q_1$; SSC = $3600/W_{net}$. Assertion–Reason: (A) The Rankine cycle has a high work ratio. (R) Pumping a liquid needs very little work. — Both true; R explains A.

30 Most Important One-Liner Facts — Rankine Cycle

1. Ideal cycle of steam power plant?

— Rankine.

2. Process in turbine?

— Isentropic expansion.

3. Process in condenser?

— Constant-pressure heat rejection.

4. Process in pump?

— Isentropic compression.

5. Process in boiler?

— Constant-pressure heat addition.

6. Turbine work?

— $h_1 - h_2$.

7. Pump work?

— $v_f(p_1 - p_2)$.

8. Why pump work small?

— Low specific volume of liquid.

9. Rankine efficiency (pump neglected)?

— $(h_1 - h_2)/(h_1 - h_{f2})$.

10. Work ratio?

— Net work ÷ turbine work.

11. Work ratio of Rankine?

— ≈ 1.

12. SSC formula?

— $3600/W_{net}$.

13. Unit of SSC?

— kg/kWh.

14. Efficiency with higher boiler pressure?

— Increases.

15. Efficiency with lower condenser pressure?

— Increases.

16. Drawback of lower condenser pressure?

— Wetter exit steam.

17. Main purpose of reheat?

— Improve exit dryness.

18. In regenerative heating?

— Feed water heated by bled steam.

19. Permissible exit dryness?

— ≥ 0.88.

20. Why Carnot vapour cycle impractical?

— Compression of wet steam.

21. Rankine vs Carnot efficiency?

— Rankine lower.

22. Cycle of steam engine?

— Modified Rankine.

23. Heat supplied (with pump)?

— $h_1 - h_4$.

24. Exit dryness formula?

— $(s_1 - s_{f2})/s_{fg2}$.

25. v_f of water approx.?

— 0.001 m^3/kg.

26. Harm of wet exit steam?

— Blade erosion.

27. Effect of superheating?

— Both efficiency and dryness rise.

28. Components of Rankine plant?

— Boiler, turbine, condenser, pump.

29. Bleeding in?

— Regenerative cycle.

30. When is pump work neglected?

— At low pressures.

Practice MCQs — Rankine Cycle

Q1. Ideal cycle for a steam power plant:

(a) Otto
(b) Diesel
(c) Rankine
(d) Brayton

✔ **Answer: (c) Rankine**

Explanation: Rankine.

Q2. Pump work in the Rankine cycle:

(a) $v_f(p_1 - p_2)$
(b) $h_1 - h_2$
(c) $c_p\Delta T$
(d) Zero

✔ **Answer: (a) $v_f(p_1 - p_2)$**

Explanation: Liquid pumping.

Q3. Rankine efficiency increases with:

(a) Raising condenser pressure
(b) Raising boiler pressure
(c) Reducing superheat
(d) Removing feed heating

✔ **Answer: (b) Raising boiler pressure**

Explanation: Mean temperature of heat addition rises.

Q4. Work ratio of the Rankine cycle is:

(a) Very low
(b) Close to 1
(c) Zero
(d) 2

✔ **Answer: (b) Close to 1**

Explanation: Low pump work.

Q5. $h_1 = 3000$, $h_2 = 2100$, $h_{f2} = 300$ kJ/kg (pump neglected) — efficiency:

(a) 30%
(b) 33.3%
(c) 40%
(d) 70%

✔ **Answer: (b) 33.3%**

Explanation: 900/2700.

Q6. Specific steam consumption =

(a) $W_{net}/3600$
(b) $3600/W_{net}$
(c) $3600 \times W_{net}$
(d) Q_1/W_{net}

✔ **Answer: (b) $3600/W_{net}$**

Explanation: kg/kWh.

Q7. Main purpose of reheating:

(a) Greatly raise efficiency
(b) Increase turbine exit dryness
(c) Reduce pump work
(d) Lower boiler pressure

✔ **Answer: (b) Increase turbine exit dryness**

Explanation: Prevent erosion.

Q8. Expansion in the turbine of a Rankine cycle is assumed:

(a) Isothermal
(b) Isentropic
(c) Isobaric
(d) Throttling

✔ **Answer: (b) Isentropic**

Explanation: Reversible adiabatic.

Q9. In a regenerative cycle feed water is heated by:

(a) Exhaust gas
(b) Steam bled from turbine
(c) Electric heater
(d) Solar energy

✔ **Answer: (b) Steam bled from turbine**

Explanation: Bleeding.

Q10. Lowering condenser pressure makes exit steam:

(a) Drier
(b) Wetter
(c) Superheated
(d) Unchanged

✔ **Answer: (b) Wetter**

Explanation: More expansion.

Q11. Rankine compared to Carnot (same limits):

(a) More efficient
(b) Less efficient, more practical
(c) Equal
(d) Impractical

✔ **Answer: (b) Less efficient, more practical**

Explanation: No wet-steam compression.

Q12. W_{net} = 1000 kJ/kg — SSC:

(a) 1 kg/kWh
(b) 3.6 kg/kWh
(c) 0.36 kg/kWh
(d) 36 kg/kWh

✔ **Answer: (b) 3.6 kg/kWh**

Explanation: 3600/1000.

Q13. Heat addition in the boiler is at constant:

(a) Volume
(b) Pressure
(c) Temperature
(d) Entropy

✔ **Answer: (b) Pressure**

Explanation: Constant pressure.

Q14. Pump work is usually ____ of turbine work:

(a) 1–2%
(b) 20%
(c) 50%
(d) 90%

✔ **Answer: (a) 1–2%**

Explanation: Very small.

Q15. $s_1 = 6.6$, $s_{f2} = 0.6$, $s_{fg2} = 7.5$ — x_2:

(a) 0.70
(b) 0.80
(c) 0.88
(d) 0.95

✔ **Answer: (b) 0.80**

Explanation: 6/7.5.

SSC JE / UPSSSC JE Previous Year Questions — Rankine Cycle

Q1. The Rankine cycle consists of: SSC JE (Previous Year)

(a) Two isentropic and two constant pressure processes
(b) Two isothermal and two isentropic
(c) Two constant volume and two isentropic
(d) Four isothermal

✔ **Answer: (a) Two isentropic and two constant pressure processes**

Explanation: Rankine.

Q2. Rankine efficiency can be increased by: SSC JE (Previous Year)

(a) Decreasing condenser pressure
(b) Decreasing boiler pressure
(c) Decreasing superheat
(d) All of these

✔ **Answer: (a) Decreasing condenser pressure**

Explanation: Lower condenser pressure.

Q3. Regenerative heating: SSC JE (Previous Year)

(a) Increases efficiency
(b) Decreases efficiency
(c) Increases work
(d) Has no effect

✔ **Answer: (a) Increases efficiency**

Explanation: Mean heat-addition temperature rises.

Q4. Pump work in Rankine is negligible because: SSC JE (Previous Year)

(a) Low specific volume of liquid
(b) Small pressure difference
(c) Efficient pump
(d) High temperature

✔ **Answer: (a) Low specific volume of liquid**

Explanation: v_f is very small.

Q5. In a reheat cycle steam is reheated:

SSC JE (Previous Year)

(a) Before boiler

(b) After partial expansion in turbine

(c) After condenser

(d) After pump

✔ **Answer: (b) After partial expansion in turbine**

Explanation: After the HP turbine.

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Chapter 2: Boilers — Classification, Specifications, Mountings & Accessories

Boilers — Classification, Specifications, Mountings & Accessories – Basic Concept

A **boiler (steam generator)** is a closed vessel in which water is converted into steam using the heat of combustion of fuel. Under the Indian Boilers Act 1923, a closed vessel of capacity over 22.75 litres generating steam above 1 kg/cm² (gauge) is a boiler. Boilers are used for power generation, industry (textile, sugar, paper) and heating.

Understand in Detail — Boilers — Classification, Specifications, Mountings & Accessories

2.1 Boiler (steam generator) — definition and classification

A **boiler** is a closed vessel in which water is converted into steam using heat from burning fuel. Under the Indian Boilers Act 1923, any closed vessel of 22.75 litres or more capacity in which steam is generated is a boiler — its registration and inspection are compulsory, because its bursting at high pressure can cause destruction like a bomb.

(1) Fire-tube vs water-tube: in a fire-tube boiler the hot gases flow **inside the tubes** and water is outside (in the shell) — e.g. Cochran, Lancashire, Cornish, locomotive. In a water-tube boiler water flows **inside the tubes** and hot gases outside — e.g. Babcock & Wilcox, Stirling, La Mont, Benson. **Rule:** fire-tube → low pressure (up to ≈ 25 bar), small capacity, cheap; water-tube → high pressure, large capacity, quick steaming, used in power stations.

(2) Position of shell: vertical (Cochran, simple vertical) and horizontal (Lancashire, locomotive, B&W). **(3) Position of furnace:** internally fired (Cochran, Lancashire, locomotive) and externally fired (B&W). **(4) Circulation: natural** — hot water and steam become lighter and rise, cold water comes down (B&W, Lancashire); **forced** — by a pump (La Mont, Benson, Loeffler). **(5) Pressure:** low, medium, high and **supercritical** — above 221.2 bar, where water turns into steam without boiling (Benson).

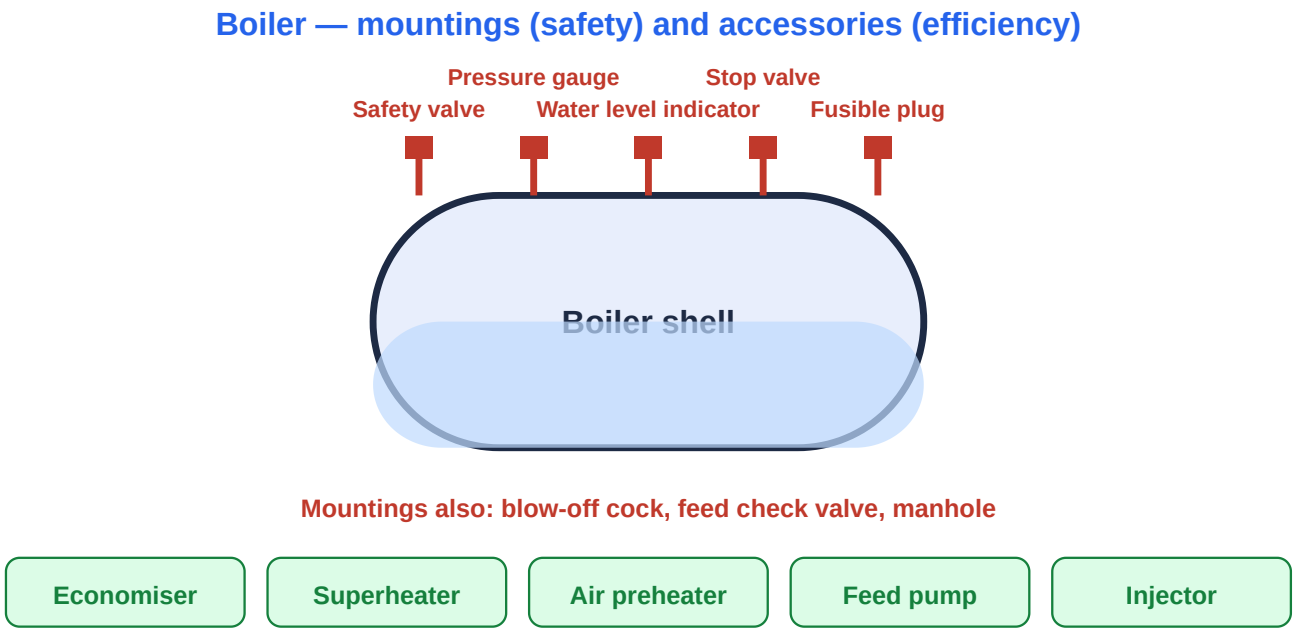


Fig 6.2: Boiler mountings and accessories

2.2 Identifying major boilers — for the exam

Cochran: vertical, multi-tubular, fire-tube, internally fired; hemispherical dome-like top; small space, small industries.

Lancashire: horizontal, **two** large fire tubes (Cornish has one); gases pass first through the tubes, then the bottom flue, then the side flues — three passes; heavy, slow but reliable; sugar mills, chemical industries.

Locomotive: horizontal, multi-tubular, fire-tube; much steam in little time; **artificial draught** — exhaust steam blown into the chimney through a blast pipe. Used in steam railway engines.

Babcock & Wilcox (B&W): horizontal, water-tube, externally fired, natural circulation; tubes **inclined at 15°** for good circulation; high capacity. **La Mont:** forced circulation, water-tube. **Benson: drumless**, supercritical pressure, once-through. **Loeffler:** no water flows through the tubes — superheated steam is bubbled through water to generate steam, so no scale forms in tubes. **Velox:** very high gas velocity (supersonic), extremely rapid heat transfer.

2.3 Mountings and accessories — the difference and their jobs

Basic difference: mountings = devices essential for the **safety and control** of a boiler — running a boiler without them is legally prohibited. **Accessories** = auxiliary devices that **increase efficiency** — a boiler can run without them, but at lower efficiency.

Mountings (with functions): (1) **safety valve** — releases steam as soon as pressure exceeds the limit; types: dead-weight, lever, spring-loaded (locomotive/marine — works even with vibration), **high-steam and low-water safety valve** (protects against both dangers). (2) **water level indicator** — water level in a glass tube. (3) **pressure gauge** — Bourdon tube. (4) **fusible plug** — above the furnace, filled with low-melting metal (tin/lead); if the water level falls and exposes the furnace crown, the metal melts and

steam-water falls and puts out the fire — **protects the furnace from burning**. (5) **stop valve** — starts/stops steam flow from the boiler. (6) **feed check valve** — lets water into the boiler, not back. (7) **blow-off cock** — removes mud/sediment from the bottom and empties the boiler. (8) **manhole** — for inspection and cleaning.

Accessories (in order along the gas path): (1) **superheater** — makes saturated steam superheated (raises temperature, not pressure). (2) **economiser** — heats the **feed water** with leftover heat of flue gases. (3) **air preheater** — heats the **air** going to the furnace using flue gases (after the economiser, before the chimney). (4) **feed pump** and **injector** (feeding water by steam jet). (5) **steam separator/trap**.

> **Note:** Memory trick — 'safety = mounting, saving = accessory'. Fusible plug, safety valve, water level indicator, pressure gauge → mountings; economiser, superheater, air preheater, feed pump → accessories.

2.4 Boiler specifications, equivalent evaporation and efficiency

How do we compare two boilers when one makes steam at 10 bar from feed water at 50°C and the other at 15 bar from 80°C? Comparing 'kg of steam/hour' directly would be unfair, as the heat used per kg differs. That is why a standard has been fixed.

Equivalent evaporation ('from and at 100°C'): how much water at 100°C could be converted into dry saturated steam at 100°C using the total heat given in the actual evaporation. The latent heat of water at 100°C is 2257 kJ/kg, so $\mathbf{E = m_a(h - h_f)/2257}$, where m_a = actual evaporation, h = enthalpy of steam produced, h_f = enthalpy of feed water. The ratio $(h - h_f)/2257$ is called the **factor of evaporation**.

Boiler efficiency = heat used in producing steam ÷ heat supplied by fuel = $\mathbf{m_s(h - h_f)/(m_f \times CV)}$, where m_s = steam/hour, m_f = fuel/hour, CV = calorific value of fuel. A **boiler specification** states: type, working pressure, steam output (kg/h), heating surface (m^2), fuel, efficiency, equivalent evaporation.

Draught: the pressure difference needed to supply air to the furnace and push gases out through the chimney. **Natural draught** — from the density difference between hot light chimney gases and cold heavy outside air (in mm of water): $h = 353H(1/T_a - 1/T_g)$. **Artificial** — a **forced** fan pushes air before the furnace; an **induced** fan pulls gases near the chimney; **balanced** — both.

> **Example:** A boiler produces 8 kg of steam per kg of coal; h = 2780 kJ/kg, feed water at 40°C (h_f = 167.5). CV of coal = 30000 kJ/kg. Find efficiency and equivalent evaporation.
>
> **Solution:**
>
> 1. Heat per kg of steam = 2780 − 167.5 = 2612.5 kJ
> 2. $\eta = 8 \times 2612.5/30000 =$ **69.7%**
> 3. $E = 8 \times 2612.5/2257 =$ **9.26 kg per kg of coal** (from and at 100°C)

1. Key Points

- **What flows in tubes:** fire-tube — hot gases inside tubes, water outside (Cornish, Lancashire, Cochran, Locomotive); water-tube — water inside tubes, gases outside (Babcock & Wilcox, Stirling, La Mont, Benson).
- **Axis of shell:** horizontal (Lancashire, Locomotive), vertical (Cochran, simple vertical) and inclined.
- **Furnace position:** internally fired (Lancashire, Cochran) and externally fired (Babcock & Wilcox).
- **Water circulation:** natural (density difference — B&W) and forced (pump — La Mont, Benson, Velox).
- **Pressure:** low pressure (< 10 bar – Cochran, Lancashire), high pressure (B&W, La Mont), supercritical (> 221 bar — Benson; drumless).
- **Boiler mountings (for safety, compulsory):** safety valves (2), water level indicators (2), pressure gauge, fusible plug, steam stop valve, feed check valve, blow-off cock, manhole.
- **Boiler accessories (to raise efficiency):** economiser (heats feed water by flue gas), air preheater (heats combustion air), superheater (superheats steam), feed pump, injector, steam separator, steam trap.
- **Types of safety valve:** dead weight, lever, spring loaded (locomotive/marine), and high steam–low water safety valve.
- **Fusible plug:** on the furnace crown; when water level falls, the low-melting metal melts, letting steam and water into the furnace to put out the fire.
- **Boiler specifications:** type, working pressure, steam generation capacity (kg/h), equivalent evaporation, heating surface area, efficiency, type of fuel.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Boiler efficiency	$\eta = m_s(h - h_{fw}) / (m_f \times CV)$
Equivalent evaporation	$m_e = m_s(h - h_{fw})/2257$
Factor of evaporation	$F = (h - h_{fw})/2257$
Actual evaporation	$m_a = m_s/m_f$ (kg steam/kg fuel)
Latent heat 'from and at 100°C'	2257 kJ/kg
Natural draught of chimney	$h_w = 353H(1/T_a - (m+1)/(m\ T_g))$ mm

3. Detailed Explanation

- **Cochran boiler:** vertical, multi-fire-tube, internally fired; compact, ~10 bar; for small industries.
- **Lancashire boiler:** horizontal, two large fire tubes, internally fired; large steam storage, handles sudden load changes; used in sugar mills.
- **Babcock & Wilcox boiler:** horizontal inclined water tubes, externally fired, natural circulation; high pressure and capacity — used in power plants.

- **La Mont and Benson:** forced circulation; Benson works drumless at supercritical pressure — water flashes directly into steam.
- **Economiser vs air preheater:** both recover flue-gas heat — the first heats feed water, the second heats combustion air; order: boiler → superheater → economiser → air preheater → chimney.
- **Draught:** the pressure difference that drives air through the furnace; natural (chimney) and artificial — induced (fan at chimney base), forced (fan before furnace), balanced.

4. Comparison Table — Fire-tube vs Water-tube Boiler

Basis	Fire-tube	Water-tube
Inside tubes	Hot gases	Water
Pressure limit	Low (up to ≈ 20 bar)	High (> 100 bar)
Steam rate	Low	High
Explosion risk	More (large water volume)	Less
Examples	Cochran, Lancashire, Locomotive	B&W, La Mont, Benson

4. Comparison Table — Mountings vs Accessories

Basis	Mountings	Accessories
Purpose	Safety and control	Increase efficiency
Compulsory	Yes (IBR)	No
Location	On the boiler	Separate units connected
Examples	Safety valve, fusible plug	Economiser, superheater

5. Solved Numerical Problems

Example 1. A boiler produces 80 kg of steam from 10 kg of fuel (CV = 30,000 kJ/kg); h = 2800, h_{fw} = 200 kJ/kg. Find the efficiency.

Solution:

1. $\eta = 80 \times (2800 - 200)/(10 \times 30000) = 208000/300000 =$ **69.3%**

Example 2. For the above, find the equivalent evaporation per kg of fuel.

Solution:

1. $m_a = 80/10 = 8$ kg/kg
2. $m_e = 8 \times 2600/2257 =$ **9.22 kg/kg of fuel**

Example 3. Find the factor of evaporation if h = 2900 and h_{fw} = 300 kJ/kg.

Solution:

1. $F = (2900 - 300)/2257 =$ **1.152**

Example 4. A boiler generates 5000 kg/h of steam ($h - h_{fw}$ = 2500 kJ/kg) at 80% efficiency. Find fuel (CV = 40,000) consumption.

Solution:

1. $m_f = 5000 \times 2500/(0.8 \times 40000) =$ **390.6 kg/h**

6. Frequently Asked Facts

- Fire-tube: Cochran, Lancashire, Locomotive, Cornish; water-tube: B&W, Stirling, La Mont, Benson.
- Benson — supercritical, drumless.
- Mountings for safety, accessories for efficiency.
- Equivalent evaporation 'from and at 100°C' — 2257 kJ/kg.

7. Note (Confusing Facts)

- Economiser heats feed water; air preheater heats air — both are accessories.
- Feed check valve is a mounting; feed pump is an accessory.
- A locomotive boiler is fire-tube, though it gives a high steam rate.

Remember: One line: 2 safety valves + 2 water level indicators are compulsory. Assertion–Reason: (A) Water-tube boilers suit high pressure. (R) Small-diameter tubes withstand higher pressure. — Both true; R explains A.

30 Most Important One-Liner Facts — Boilers — Classification, Specifications, Mountings & Accessories

1. Gases inside tubes — boiler?

— Fire-tube.

2. Water inside tubes — boiler?

— Water-tube.

3. Type of Cochran boiler?

— Vertical, multi-fire-tube.

4. Fire tubes in Lancashire?

— Two.

5. Fire tubes in Cornish?

— One.

6. Type of B&W boiler?

— Water-tube, externally fired.

7. Forced circulation boilers?

— La Mont, Benson, Velox.

8. Drumless boiler?

— Benson.

9. Supercritical boiler?

— Benson.

10. Locomotive boiler type?

— Horizontal fire-tube.

11. Purpose of mountings?

— Safety.

12. Purpose of accessories?

— Increase efficiency.

13. Compulsory safety valves?

— 2.

14. Compulsory water gauges?

— 2.

15. Puts off fire on low water?

— Fusible plug.

16. Location of fusible plug?

— Furnace crown.

17. Heats feed water with flue gas?

— Economiser.

18. Heats combustion air?

— Air preheater.

19. Superheats steam?

— Superheater.

20. Removes water and sludge?

— Blow-off cock.

21. Prevents water flowing back?

— Feed check valve.

22. Safety valve on locomotives?

— Spring loaded.

23. Boiler efficiency?

— $m_s(h - h_{fw})/(m_f \, CV)$.

24. Latent heat in equivalent evaporation?

— 2257 kJ/kg.

25. Year of Indian Boilers Act?

— 1923.

26. Fan at chimney base — draught?

— Induced.

27. Fan before furnace — draught?

— Forced.

28. Draught by chimney?

— Natural.

29. Removes condensate?

— Steam trap.

30. Feeds water without pump?

— Injector.

Practice MCQs — Boilers — Classification, Specifications, Mountings & Accessories

Q1. Which is a water-tube boiler?

(a) Cochran
(b) Lancashire
(c) Babcock & Wilcox
(d) Locomotive

✔ **Answer: (c) Babcock & Wilcox**

Explanation: B&W is water-tube.

Q2. Function of the fusible plug:

(a) Measure pressure
(b) Extinguish fire on low water
(c) Superheat steam
(d) Heat feed water

✔ **Answer: (b) Extinguish fire on low water**

Explanation: Safety mounting.

Q3. An economiser heats:

(a) Combustion air
(b) Feed water
(c) Steam
(d) Fuel

✔ **Answer: (b) Feed water**

Explanation: Using flue-gas heat.

Q4. Which is a boiler accessory?

(a) Safety valve
(b) Pressure gauge
(c) Superheater
(d) Fusible plug

✔ **Answer: (c) Superheater**

Explanation: For efficiency.

Q5. Drumless boiler working at supercritical pressure:

(a) La Mont
(b) Benson
(c) Lancashire
(d) Cochran

✔ **Answer: (b) Benson**

Explanation: Benson.

Q6. Number of fire tubes in a Lancashire boiler:

(a) 1
(b) 2
(c) 4
(d) Many

✔ **Answer: (b) 2**

Explanation: Two.

Q7. A locomotive boiler is:

(a) Water-tube
(b) Horizontal multi-fire-tube
(c) Vertical water-tube
(d) Forced circulation

✔ **Answer: (b) Horizontal multi-fire-tube**

Explanation: Fire-tube.

Q8. Number of compulsory safety valves on a boiler:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

Explanation: IBR.

Q9. Device that heats combustion air:

(a) Economiser
(b) Air preheater
(c) Superheater
(d) Condenser

✔ **Answer: (b) Air preheater**

Explanation: From flue gas.

Q10. To remove sludge and sediment from a boiler:

(a) Blow-off cock
(b) Stop valve
(c) Feed check valve
(d) Safety valve

✔ **Answer: (a) Blow-off cock**

Explanation: Blow-off cock.

Q11. 80 kg steam/10 kg fuel, h − h_{fw} = 2257 kJ/kg — equivalent evaporation:

(a) 8 kg/kg
(b) 10 kg/kg
(c) 2257 kg/kg
(d) 0.8 kg/kg

✔ **Answer: (a) 8 kg/kg**

Explanation: F = 1.

Q12. Natural circulation is due to:

(a) Pump
(b) Density difference
(c) Chimney
(d) Fan

✔ **Answer: (b) Density difference**

Explanation: Hot water is lighter.

Q13. Low-pressure vertical boiler for small industries:

(a) Cochran
(b) B&W
(c) Benson
(d) La Mont

✔ **Answer: (a) Cochran**

Explanation: Cochran.

Q14. Fan at the base of the chimney — draught is:

(a) Natural
(b) Forced
(c) Induced
(d) Balanced

✔ **Answer: (c) Induced**

Explanation: It sucks gases.

Q15. The feed check valve is a:

(a) Mounting
(b) Accessory
(c) Fuel valve
(d) Turbine part

✔ **Answer: (a) Mounting**

Explanation: Compulsory fitting on boiler.

SSC JE / UPSSSC JE Previous Year Questions — Boilers — Classification, Specifications, Mountings & Accessories

Q1. Which is not a boiler mounting? SSC JE (Previous Year)

(a) Safety valve
(b) Water level indicator
(c) Economiser
(d) Fusible plug

✔ **Answer: (c) Economiser**

Explanation: Economiser is an accessory.

Q2. Cochran boiler is a: SSC JE (Previous Year)

(a) Horizontal water-tube
(b) Vertical fire-tube
(c) Horizontal fire-tube
(d) Vertical water-tube

✔ **Answer: (b) Vertical fire-tube**

Explanation: Vertical multi-fire-tube.

Q3. Special feature of Benson boiler: SSC JE (Previous Year)

(a) Large drum
(b) No drum
(c) Low pressure
(d) Fire-tube

✔ **Answer: (b) No drum**

Explanation: Supercritical, drumless.

Q4. Function of a superheater: SSC JE (Previous Year)

(a) Heat feed water
(b) Raise steam temperature above saturation
(c) Heat air
(d) Increase pressure

✔ **Answer: (b) Raise steam temperature above saturation**

Explanation: Superheating.

Q5. Lancashire boiler is: SSC JE (Previous Year)

(a) Internally fired horizontal fire-tube
(b) Externally fired water-tube
(c) Vertical
(d) Forced circulation

✔ **Answer: (a) Internally fired horizontal fire-tube**

Explanation: Two fire tubes.

Q6. Protects the boiler when water level falls: SSC JE (Previous Year)

(a) Fusible plug
(b) Blow-off cock
(c) Stop valve
(d) Injector

✔ **Answer: (a) Fusible plug**

Explanation: Plug melts and quenches the fire.

UPSSSC ALERT

Chapter 3: Steam Nozzles

Steam Nozzles – Basic Concept

A **nozzle** is a passage of varying cross-section in which steam pressure drops and its enthalpy is converted into kinetic energy (high velocity). In steam turbines the high-velocity jet from nozzles strikes the blades and rotates them. Nozzles are also used in injectors and ejectors.

Understand in Detail — Steam Nozzles

3.1 Steam nozzle — the art of turning pressure into velocity

When you squeeze the mouth of a garden hose, water shoots far — because the narrower passage increases the velocity. A **nozzle** is such a passage of varying cross-section in which the **pressure (enthalpy) of steam falls and its kinetic energy rises**. In a steam turbine this fast jet strikes the blades and turns them — so the nozzle is the 'first step' of a turbine.

Types: (1) **convergent** — the passage keeps narrowing; exit pressure can fall only to the critical pressure, velocity at most sonic. (2) **divergent** — keeps widening. (3) **convergent-divergent (de Laval)** — first narrows, at the narrowest section (**throat**) the velocity equals sonic velocity, then widens to give supersonic velocity. Used when the pressure ratio is below critical.

Why does the passage widen in supersonic flow? In subsonic flow a narrowing passage increases velocity (like water). But above sonic velocity the density of steam falls so fast that to increase velocity the **area must increase** (in $\dot{m} = \rho AV$, ρ is falling very rapidly). That is why a rocket nozzle has a wide exit.

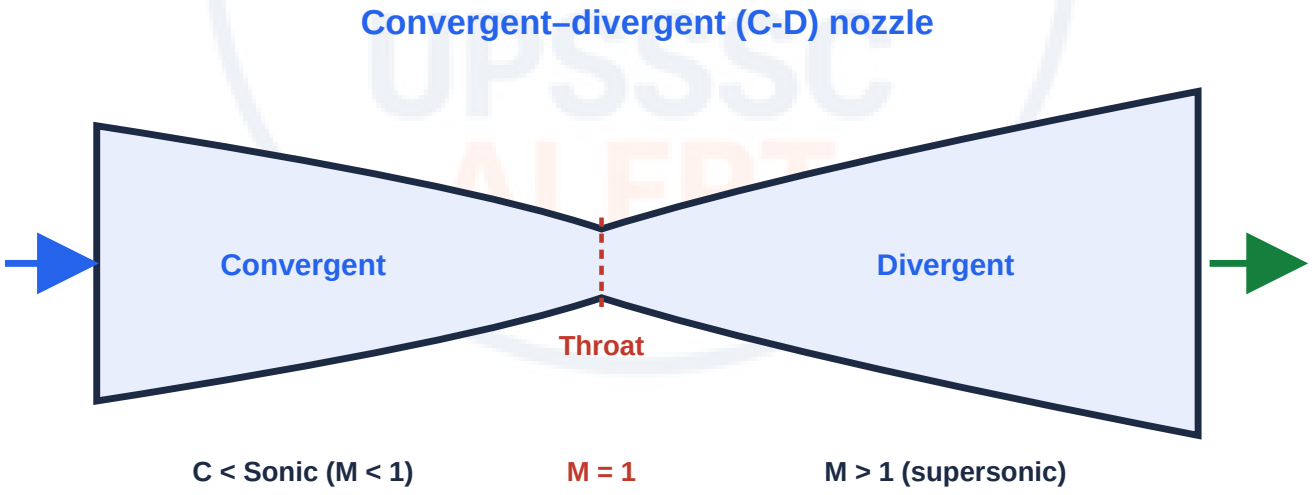


Fig 6.3: Steam nozzle — convergent–divergent

Derivation: Nozzle exit velocity

1. Steady flow energy equation (no work, no heat, negligible height difference): $h_1 + V_1^2/2 = h_2 + V_2^2/2$
2. Inlet velocity negligible ($V_1 \approx 0$): $V_2 = \sqrt{[2(h_1 - h_2)]}$ (h in J/kg)
3. Taking h in kJ/kg: $\mathbf{V_2 = 44.72\ \sqrt{(h_1 - h_2)}}$ m/s — worth memorising
4. With friction the actual heat drop = η_{nozzle} × isentropic heat drop

Critical pressure ratio: the pressure ratio p_c/p_1 at which the mass flow per unit area is **maximum** and the velocity at the throat equals sonic velocity: $\mathbf{p_c/p_1 = [2/(n + 1)]^{n/(n-1)}}$. Dry saturated steam (n = 1.135) → **0.577**; superheated steam (n = 1.3) → **0.546**; air (n = 1.4) → **0.528**.

Choked flow: once critical conditions are reached at the throat, further lowering of the exit pressure does not increase the mass flow — the nozzle is 'choked'. **Supersaturated (metastable) flow:** in rapid expansion the steam has no time to condense and stays dry even after crossing the saturation line (up to the Wilson line) — this slightly increases the mass flow and slightly reduces the heat drop.

Example: Steam enters a nozzle at 10 bar, h_1 = 2778 kJ/kg; at exit h_2 = 2550 kJ/kg (isentropic). Inlet velocity negligible, nozzle efficiency 90%. Exit velocity?

Solution:

1. Isentropic heat drop = 2778 − 2550 = 228 kJ/kg
2. Actual = 0.9 × 228 = 205.2 kJ/kg
3. $V_2 = 44.72\ \sqrt{205.2} = 44.72 \times 14.32 =$ **640.6 m/s**
4. Critical pressure (dry saturated) = 0.577 × 10 = **5.77 bar**

1. Key Points

- **Types of nozzles:** convergent (up to sonic velocity), divergent, and convergent–divergent (C-D / de Laval – for supersonic velocity).
- **Exit velocity (from SFEE):** $C_2 = \sqrt{[2(h_1 - h_2) + C_1^2]} \approx 44.72\sqrt{(h_1 - h_2)}$ m/s (h in kJ/kg, C_1 negligible).
- **Throat:** the section of minimum area in a C-D nozzle; at maximum flow the velocity there is sonic (M = 1).
- **Critical pressure ratio:** $p_c/p_1 = [2/(n + 1)]^{n/(n-1)}$; superheated steam (n = 1.3) 0.546, saturated steam (n = 1.135) 0.577, air (n = 1.4) 0.528.
- **Choked flow:** reducing the back pressure below critical does not increase mass flow in a convergent nozzle; it stays at its maximum.
- **Nozzle efficiency:** η_n = actual enthalpy drop ÷ isentropic enthalpy drop; friction reduces the actual velocity. Velocity coefficient = $\sqrt{\eta_n}$.
- **Supersaturated (metastable) flow:** in rapid expansion steam crosses the saturation line without condensing for a while, up to the Wilson line. This slightly increases mass flow.

- **Mass flow rate:** m = A·C/v, where v is the specific volume of steam at that section.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Exit velocity	$C_2 = 44.72\sqrt{(h_1 - h_2)}$
With inlet velocity	$C_2 = \sqrt{[2000(h_1 - h_2) + C_1^2]}$
Critical pressure ratio	$[2/(n+1)]^{n/(n-1)}$
Superheated steam	$n = 1.3 \rightarrow 0.546$
Saturated steam	$n = 1.135 \rightarrow 0.577$
Air	$n = 1.4 \rightarrow 0.528$
Mass flow	$m = A\,C / v$
Nozzle efficiency	$\eta_n = \Delta h_{actual}/\Delta h_{isentropic}$
Velocity coefficient	$C_{actual}/C_{isentropic} = \sqrt{\eta_n}$

3. Detailed Explanation

- **Area–velocity relation:** dA/A = (M² − 1) dC/C. In subsonic flow (M < 1) area must decrease to accelerate (convergent); in supersonic flow (M > 1) area must increase (divergent).
- **Use of C-D nozzle:** when the exit pressure is below the critical pressure (large pressure ratio), for supersonic velocity; e.g. de Laval turbines and rockets.
- **Effect of friction in nozzle:** actual enthalpy drop decreases, exit steam is drier (reheat effect), entropy increases.
- **Index n of steam expansion:** $pv^n = C$; superheated 1.3, saturated 1.135, wet steam n = 1.035 + 0.1x (Zeuner).

4. Comparison Table — Convergent vs C-D Nozzle

Basis	Convergent	Convergent–divergent
Max exit velocity	Sonic (M = 1)	Supersonic (M > 1)
Use	Small pressure ratio	Large pressure ratio
Throat	Exit is the throat	Throat in the middle
Example	Impulse stages	De Laval turbine, rocket

5. Solved Numerical Problems

Example 1. Isentropic enthalpy drop in a steam nozzle is 200 kJ/kg, inlet velocity negligible. Find the exit velocity.

Solution:

1. $C_2 = 44.72\sqrt{200} = 44.72 \times 14.14 =$ **632.5 m/s**

Example 2. If the above nozzle efficiency is 90%, find the actual velocity.

Solution:

1. $\Delta h = 0.9 \times 200 = 180$ kJ/kg
2. $C = 44.72 \sqrt{180} =$ **600 m/s**

Example 3. Superheated steam enters a nozzle at 10 bar. Find the throat pressure.

Solution:

1. $p_c = 0.546 \times 10 =$ **5.46 bar**

Example 4. At exit C = 500 m/s, v = 0.5 m³/kg and flow 2 kg/s. Find the exit area.

Solution:

1. $A = m\, v/C = 2 \times 0.5/500 = 0.002$ m² = **2000 mm²**

6. Frequently Asked Facts

- $C = 44.72\sqrt{\Delta h}$ — the most asked formula.
- Critical pressure ratio: superheated 0.546, saturated 0.577, air 0.528.
- M = 1 at the throat; C-D nozzle for supersonic flow.
- Friction in a nozzle makes exit steam drier.

7. Note (Confusing Facts)

- In subsonic flow the convergent part accelerates; in supersonic flow the divergent part does.
- After choking, lowering back pressure does not increase flow.
- The factor 44.72 applies only when Δh is in kJ/kg.

Remember: One line: nozzle – enthalpy → kinetic energy; diffuser – the reverse. Assertion–Reason: (A) A convergent nozzle cannot give supersonic velocity. (R) The maximum velocity at the throat is sonic. — Both true; R explains A.

30 Most Important One-Liner Facts — Steam Nozzles

1. Function of nozzle?

— Convert enthalpy into KE.

2. Function of diffuser?

— Convert KE into pressure.

3. Exit velocity formula?

— $44.72\sqrt{\Delta h}$.

4. Nozzle for supersonic?

— Convergent–divergent.

5. Other name of C-D nozzle?

— De Laval nozzle.

6. Section of minimum area?

— Throat.

7. Mach number at throat?

— 1.

8. Critical ratio for superheated?

— 0.546.

9. Critical ratio for saturated?

— 0.577.

10. Critical ratio for air?

— 0.528.

11. n for superheated steam?

— 1.3.

12. n for saturated steam?

— 1.135.

13. n for wet steam (Zeuner)?

— 1.035 + 0.1x.

14. Mass flow?

— AC/v.

15. Nozzle efficiency?

— Actual ÷ isentropic enthalpy drop.

16. Velocity coefficient?

— $\sqrt{\eta_n}$.

17. Area to accelerate subsonic flow?

— Decrease.

18. Area to accelerate supersonic flow?

— Increase.

19. Condition of maximum flow?

— Choked.

20. Limit of supersaturated flow?

— Wilson line.

21. Mass flow with supersaturation?

— Increases slightly.

22. Exit steam with friction?

— Drier.

23. Entropy with friction?

— Increases.

24. Area–velocity relation?

— $dA/A = (M^2 - 1)dC/C$.

25. Max velocity in convergent nozzle?

— Sonic velocity.

26. Flow in nozzle assumed?

— Adiabatic.

27. Work in nozzle?

— Zero.

28. Velocity for Δh = 125 kJ/kg?

— 500 m/s.

29. Nozzle in rocket?

— Convergent–divergent.

30. Throat pressure for 10 bar superheated?

— 5.46 bar.

Practice MCQs — Steam Nozzles

Q1. Exit velocity of steam in a nozzle (Δh in kJ/kg):

(a) $4.472\sqrt{\Delta h}$ (b) $44.72\sqrt{\Delta h}$

(c) $447.2\sqrt{\Delta h}$ (d) $91.5\sqrt{\Delta h}$

✔ **Answer: (b) $44.72\sqrt{\Delta h}$**

Explanation: $\sqrt{(2000\Delta h)}$.

Q2. Critical pressure ratio for superheated steam:

(a) 0.528 (b) 0.546

(c) 0.577 (d) 0.6

✔ **Answer: (b) 0.546**

Explanation: n = 1.3.

Q3. Nozzle used for supersonic velocity:

(a) Convergent
(b) Divergent
(c) Convergent–divergent
(d) Parallel

✔ **Answer: (c) Convergent–divergent**

Explanation: Divergent after throat.

Q4. Mach number at the throat (maximum flow):

(a) 0.5
(b) 1
(c) 2
(d) Zero

✔ **Answer: (b) 1**

Explanation: Sonic.

Q5. Value of n for saturated steam:

(a) 1.3
(b) 1.4
(c) 1.135
(d) 1.035

✔ **Answer: (c) 1.135**

Explanation: 1.135.

Q6. Δh = 180 kJ/kg — exit velocity:

(a) 420 m/s
(b) 600 m/s
(c) 800 m/s
(d) 180 m/s

✔ **Answer: (b) 600 m/s**

Explanation: 44.72 × 13.42.

Q7. Friction in a nozzle makes exit steam:

(a) Wetter
(b) Drier
(c) Unchanged
(d) Subcooled

✔ **Answer: (b) Drier**

Explanation: Reheat effect.

Q8. To accelerate subsonic flow, the nozzle must be:

(a) Convergent
(b) Divergent
(c) Parallel
(d) Any

✔ **Answer: (a) Convergent**

Explanation: dA < 0.

Q9. After choking, reducing back pressure makes the flow:

(a) Increase
(b) Decrease
(c) Remain constant
(d) Zero

✔ **Answer: (c) Remain constant**

Explanation: Maximum flow.

Q10. Limit of supersaturated flow:

(a) Wilson line
(b) Saturation line
(c) Critical point
(d) Fanno line

✔ **Answer: (a) Wilson line**

Explanation: Wilson line.

Q11. Nozzle efficiency 81% — velocity coefficient:

(a) 0.81
(b) 0.9
(c) 0.66
(d) 0.95

✔ **Answer: (b) 0.9**

Explanation: √0.81.

Q12. A de Laval nozzle is:

(a) Convergent
(b) Convergent–divergent
(c) Divergent
(d) Diffuser

✔ **Answer: (b) Convergent–divergent**

Explanation: C-D.

Q13. Steam flow in a nozzle is assumed:

(a) Isothermal
(b) Adiabatic
(c) Constant pressure
(d) Constant volume

✔ **Answer: (b) Adiabatic**

Explanation: Negligible heat transfer.

Q14. Critical pressure ratio for air:

(a) 0.528
(b) 0.546
(c) 0.577
(d) 0.5

✔ **Answer: (a) 0.528**

Explanation: n = 1.4.

Q15. 2 kg/s, v = 0.5 m^3/kg, C = 500 m/s — exit area:

(a) 200 mm^2
(b) 2000 mm^2
(c) 20000 mm^2
(d) 500 mm^2

✔ **Answer: (b) 2000 mm^2**

Explanation: mv/C = 0.002 m^2.

SSC JE / UPSSSC JE Previous Year Questions — Steam Nozzles

Q1. Purpose of a steam nozzle: SSC JE (Previous Year)

(a) Increase pressure
(b) Convert heat energy into kinetic energy
(c) Condense steam
(d) Superheat steam

✔ **Answer: (b) Convert heat energy into kinetic energy**

Explanation: Enthalpy → velocity.

Q2. Critical pressure ratio for dry saturated steam: SSC JE (Previous Year)

(a) 0.528
(b) 0.546
(c) 0.577
(d) 0.582

✔ **Answer: (c) 0.577**

Explanation: 0.577.

Q3. Velocity in the divergent part of a C-D nozzle: SSC JE (Previous Year)

(a) Decreases
(b) Increases (supersonic)
(c) Remains constant
(d) Zero

✔ **Answer: (b) Increases (supersonic)**

Explanation: M > 1.

Q4. Exit velocity of steam from a nozzle depends on: SSC JE (Previous Year)

(a) Enthalpy drop
(b) Nozzle length
(c) Colour of steam
(d) Mass

✔ **Answer: (a) Enthalpy drop**

Explanation: $C \propto \sqrt{\Delta h}$.

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Chapter 4: Steam Turbines

Steam Turbines – Basic Concept

A **steam turbine** is a rotary machine in which the heat energy of steam is first converted into kinetic energy in nozzles/blades and then into rotary mechanical work. It is the main prime mover of thermal and nuclear power plants, with large capacity, high speed and balanced motion.

Understand in Detail — Steam Turbines

4.1 Steam turbine — the principle of impulse and reaction

To understand **impulse**, think — you stand on a cart and catch a fast-coming ball, and the cart moves back. In an **impulse turbine** the entire pressure drop of the steam takes place only in the **nozzles**; the fast jet strikes the moving blades and the change in direction (change of momentum) produces force. The passage between blades has constant area, so **pressure stays constant across the moving blades**.

To understand **reaction** — release an inflated balloon and it flies opposite to the escaping air. In a **reaction turbine** the passage between the moving blades is also **nozzle-shaped** (convergent); steam expands in the moving blades too, its relative velocity increases, and the reaction pushes the blades. Here the **pressure drop occurs in both fixed and moving blades**. The practical reaction turbine = **Parsons turbine** (50% reaction).

Degree of reaction R = enthalpy drop in moving blades ÷ enthalpy drop in the whole stage. Impulse turbine: R = 0; Parsons: R = 0.5 (fixed and moving blades have the same shape — **symmetrical blades**).

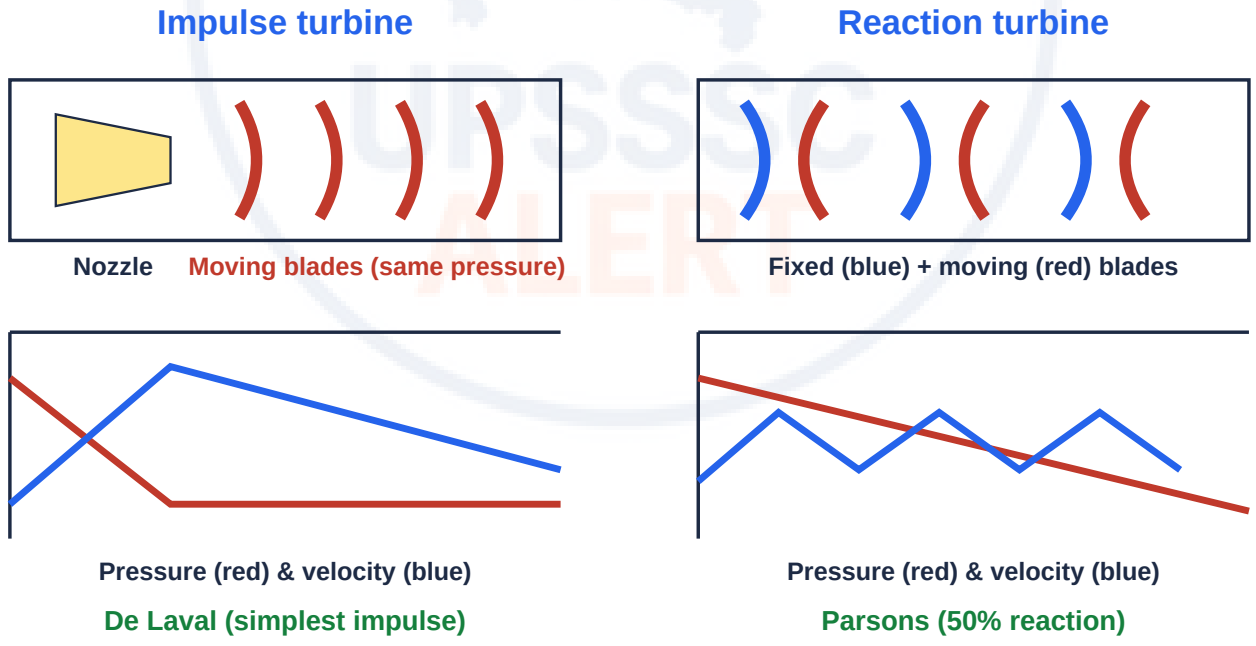


Fig 6.4: Pressure and velocity variation in impulse and reaction turbines

4.2 Velocity triangles, blade work and blade efficiency

To know how much work is done on a blade we must view velocities from two viewpoints: that of a person standing on the ground (**absolute velocity V**) and that of a person sitting on the blade (**relative velocity V_r**). Blade velocity u. They are related by the vector sum $V = u + V_r$ — drawn as a triangle. At inlet: nozzle angle α, blade inlet angle β_1; at outlet: blade angle β_2.

Whirl velocity (V_w) = component of absolute velocity along the blade motion — this produces the force. **Flow velocity (V_f)** = axial component — it carries steam forward and causes axial thrust.

Derivation: Maximum blade efficiency of an impulse turbine

1. Force on blade (per kg/s) = $V_{w1} + V_{w2}$ (added if outlet whirl is in the opposite direction) ; work = $(V_{w1} + V_{w2})\, u$
2. **Blade (diagram) efficiency $\eta_b = 2u(V_{w1} + V_{w2})/V_1^2$**
3. With symmetrical blades ($\beta_1 = \beta_2$) and no friction: $\eta_b = 4\rho(\cos\alpha - \rho)$, where $\rho = u/V_1$ (blade speed ratio)
4. $d\eta/d\rho = 0 \rightarrow$ **$\rho_{opt} = \cos\alpha / 2 \rightarrow \eta_{max} = \cos^2\alpha$**
5. For Parsons (50% reaction): **$\rho_{opt} = \cos\alpha \rightarrow \eta_{max} = 2\cos^2\alpha/(1 + \cos^2\alpha)$**

Conclusion: for maximum efficiency a reaction turbine needs **twice** the blade speed of an impulse turbine ($\cos\alpha$ vs $\cos\alpha/2$). At $\alpha = 20°$ the impulse $\eta_{max} = \cos^2 20° = 88.3\%$.

Example: In an impulse turbine the nozzle angle is 20° and steam velocity 600 m/s. Blade velocity and efficiency for maximum efficiency?

Solution:

1. $u_{opt} = V_1 \cos\alpha/2 = 600 \times 0.9397/2 =$ **281.9 m/s**
2. $\eta_{max} = \cos^2 20° = 0.9397^2 =$ **88.3%**
3. If it were Parsons, $u_{opt} = 600 \times 0.9397 = 563.8$ m/s

4.3 Compounding — taming the speed of a turbine

If steam were expanded from boiler to condenser pressure in a single stage, the jet velocity would be about 1200–1500 m/s and the blade speed for maximum efficiency half of that — at such high speed (≈ 30,000 rpm, as in the **de Laval** turbine) there are problems of centrifugal stress, vibration and gearing. So the energy is **divided into several stages** — this is called **compounding**.

(1) Velocity compounding (Curtis): pressure drops only once in the nozzle; then the velocity of the fast steam is reduced little by little in several rows of moving blades, with fixed **guide blades** between them turning the steam. Compact, but more friction and lower efficiency. **(2) Pressure compounding (Rateau):** several pairs (stages) of nozzle + moving blades; a small pressure drop in each stage's nozzles keeps the velocity low. Higher efficiency but a longer machine. **(3) Pressure-velocity compounding** — a combination of both.

Control and losses: to keep speed constant with load: **throttle governing** (reducing steam pressure), **nozzle governing** (shutting off some nozzles) or **by-pass governing**. Losses: nozzle friction, blade friction, residual velocity (kinetic energy of leaving steam), disc friction, leakage, moisture loss. **A reaction turbine has larger axial thrust** (pressure difference across the blades), balanced by a **dummy piston** or thrust bearing; and **labyrinth packing** prevents leakage at blade tips.

> **Note:** Comparison in one line: impulse — pressure drop only in nozzles, constant pressure across blades, symmetrical blade profile, partial admission possible, fewer stages, higher speed; reaction — pressure drop in both blade rows, aerofoil blades, full admission needed, more stages, more axial thrust, higher efficiency.

1. Key Points

- **Impulse turbine:** the entire pressure drop occurs in nozzles only; pressure stays constant over moving blades, only velocity falls. Examples: De Laval (simplest, single stage), Curtis, Rateau.
- **Reaction turbine:** pressure drop in both fixed and moving blades; moving blades also act as nozzles. Example: Parsons (50% reaction).
- **Degree of reaction (R):** enthalpy drop in moving blades ÷ enthalpy drop in the whole stage. Impulse R = 0; Parsons R = 0.5.
- **Compounding:** to reduce the excessive speed of a single stage (De Laval ~30,000 rpm) — velocity compounding (Curtis), pressure compounding (Rateau), pressure–velocity compounding.
- **Blade efficiency (impulse):** maximum when blade speed ratio $\rho = u/C_1 = \cos \alpha/2$; then $\eta_{max} = \cos^2\alpha$ (symmetric blades, frictionless).
- **Parsons turbine:** maximum efficiency at $u/C_1 = \cos \alpha$; $\eta_{max} = 2\cos^2\alpha/(1 + \cos^2\alpha)$.
- **Velocity diagram:** absolute velocity C, blade velocity u, relative velocity V; work from whirl components: $W = u(C_{w1} \pm C_{w2})$ per kg.
- **Turbine governing:** throttle governing, nozzle control governing, bypass governing.
- **By flow:** axial (common) and radial (Ljungström). By exhaust: condensing and back-pressure turbines.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Work per kg	$W = u\,(C_{w1} \pm C_{w2})$
Blade efficiency	$\eta_b = 2u(C_{w1} \pm C_{w2})/C_1^2$
Impulse: optimum speed ratio	$u/C_1 = \cos\alpha / 2$
Impulse: max efficiency	$\cos^2\alpha$
Parsons: optimum ratio	$u/C_1 = \cos\alpha$
Parsons: max efficiency	$2\cos^2\alpha/(1 + \cos^2\alpha)$
Degree of reaction	$R = \Delta h_{moving} / \Delta h_{stage}$
Blade speed	$u = \pi DN/60$
Axial thrust	$F_a = m\,(C_{f1} - C_{f2})$

3. Detailed Explanation

- **De Laval turbine:** single-stage impulse; very high speed (≈ 30,000 rpm), so speed must be reduced by gearing; for small outputs.
- **Velocity compounding (Curtis):** one set of nozzles followed by moving–fixed–moving blade rows; pressure drops once, velocity absorbed in stages.
- **Pressure compounding (Rateau):** several impulse stages in series, each with nozzles and blades; pressure drops in steps.
- **Axial thrust in reaction turbines:** the pressure difference across blades causes large axial thrust; balanced by a dummy piston and thrust bearing.
- **Turbine vs steam engine:** turbine — rotary, high speed, balanced, large capacity, little lubrication; steam engine — reciprocating, slow, unbalanced forces.

4. Comparison Table — Impulse vs Reaction Turbine

Basis	Impulse	Reaction
Pressure drop	Only in nozzles	In both fixed and moving blades
Pressure over moving blades	Constant	Decreases
Blade profile	Symmetrical	Aerofoil, asymmetric
Degree of reaction	0	0.5 (Parsons)
Axial thrust	Small	Large
Examples	De Laval, Curtis, Rateau	Parsons

5. Solved Numerical Problems

Example 1. In an impulse turbine the nozzle angle $\alpha = 20°$. Find the maximum blade efficiency and optimum speed ratio.

Solution:

1. $\eta_{max} = \cos^2 20° = 0.9397^2 =$ **88.3%**
2. $u/C_1 = \cos 20°/2 =$ **0.47**

Example 2. $C_1 = 600$ m/s, $\alpha = 20°$ — find the optimum blade speed of an impulse turbine.

Solution:

1. $u = C_1 \cos \alpha/2 = 600 \times 0.9397/2 =$ **281.9 m/s**

Example 3. Turbine wheel diameter 1 m and speed 3000 rpm. Find the blade speed.

Solution:

1. $u = \pi \times 1 \times 3000/60 =$ **157 m/s**

Example 4. In a stage $u = 150$ m/s, $C_{w1} + C_{w2} = 400$ m/s, flow 5 kg/s. Find the power.

Solution:

1. $W = 150 \times 400 = 60{,}000$ J/kg
2. $P = 5 \times 60{,}000 =$ **300 kW**

Example 5. In a Parsons turbine $\alpha = 20°$. Find the maximum blade efficiency.

Solution:

1. $\cos^2 20° = 0.883$
2. $\eta_{max} = 2 \times 0.883/(1 + 0.883) =$ **93.8%**

6. Frequently Asked Facts

- Impulse: pressure drop only in nozzles; reaction: in both blade rows.
- Impulse: $u/C_1 = \cos \alpha/2$, $\eta_{max} = \cos^2\alpha$.
- Parsons – 50% reaction; De Laval – simplest impulse.
- Curtis – velocity compounding; Rateau – pressure compounding.

7. Note (Confusing Facts)

- In impulse turbines only velocity (not pressure) falls over moving blades.
- The purpose of compounding is to reduce speed (rpm), not to raise efficiency.
- Parsons' optimum ratio is $\cos \alpha$, not $\cos \alpha/2$.

Remember: One line: R = 0 impulse, R = 0.5 Parsons. Assertion–Reason: (A) Gearing is used with De Laval turbines. (R) Their rotational speed is very high. — Both true; R explains A.

30 Most Important One-Liner Facts — Steam Turbines

1. Pressure drop only in nozzles — turbine?

— Impulse.

2. Pressure drop in both blades — turbine?

— Reaction.

3. Simplest impulse turbine?

— De Laval.

4. 50% reaction turbine?

— Parsons.

5. R for impulse?

— 0.

6. Purpose of compounding?

— Reduce speed.

7. Velocity compounding?

— Curtis.

8. Pressure compounding?

— Rateau.

9. Impulse optimum ratio?

— cos α/2.

10. Impulse max efficiency?

— $\cos^2\alpha$.

11. Parsons optimum ratio?

— cos α.

12. Parsons max efficiency?

— $2\cos^2\alpha/(1 + \cos^2\alpha)$.

13. Blade speed?

— $\pi DN/60$.

14. Work per kg?

— $u(C_{w1} \pm C_{w2})$.

15. Impulse blade profile?

— Symmetrical.

16. Reaction blade profile?

— Aerofoil.

17. More axial thrust?

— Reaction turbine.

18. Balancing axial thrust?

— Dummy piston.

19. Approx. speed of De Laval?

— 30,000 rpm.

20. Radial flow steam turbine?

— Ljungström.

21. Turbine supplying exhaust to process?

— Back-pressure.

22. Types of turbine governing?

— Throttle, nozzle control, bypass.

23. Degree of reaction?

— Δh moving ÷ Δh stage.

24. Moving blades act as nozzles — turbine?

— Reaction.

25. Impulse η_{max} at α = 20°?

— ≈ 88%.

26. Prime mover of thermal plant?

— Steam turbine.

27. Pressure drop in Curtis stage?

— Only once in nozzles.

28. Symbol of relative velocity?

— V.

29. Whirl velocity?

— Component along blade motion.

30. Advantage over steam engine?

— Balanced rotary motion.

Practice MCQs — Steam Turbines

Q1. In an impulse turbine the pressure drop occurs:

(a) Only in moving blades
(b) Only in nozzles
(c) In both
(d) Nowhere

✔ **Answer: (b) Only in nozzles**

Explanation: Impulse.

Q2. Parsons turbine is a:

(a) Impulse
(b) 50% reaction
(c) Velocity compounded
(d) Radial

✔ **Answer: (b) 50% reaction**

Explanation: R = 0.5.

Q3. Maximum blade efficiency of an impulse turbine:

(a) cos α
(b) $\cos^2\alpha$
(c) cos α/2
(d) $2\cos^2\alpha/(1 + \cos^2\alpha)$

✔ **Answer: (b) $\cos^2\alpha$**

Explanation: At $u/C_1 = \cos\alpha/2$.

Q4. Curtis turbine is:

(a) Pressure compounded
(b) Velocity compounded
(c) Reaction
(d) Radial

✔ **Answer: (b) Velocity compounded**

Explanation: Curtis.

Q5. Main purpose of compounding:

(a) Increase efficiency
(b) Reduce rotor speed
(c) Increase pressure
(d) Reduce weight

✔ **Answer: (b) Reduce rotor speed**

Explanation: Avoid excessive rpm.

Q6. Optimum u/C_1 for an impulse turbine:

(a) cos α
(b) cos α/2
(c) 2cos α
(d) sin α

✔ **Answer: (b) cos α/2**

Explanation: cos α/2.

Q7. Pressure across moving blades in a reaction turbine:

(a) Constant
(b) Decreases
(c) Increases
(d) Zero

✔ **Answer: (b) Decreases**

Explanation: Blades act as nozzles.

Q8. $\alpha = 20°$, $C_1 = 600$ m/s — impulse optimum blade speed:

(a) 564 m/s
(b) 282 m/s
(c) 300 m/s
(d) 141 m/s

✔ **Answer: (b) 282 m/s**

Explanation: $600 \times 0.94/2$.

Q9. D = 1 m, N = 3000 rpm — blade speed:

(a) 50 m/s
(b) 157 m/s
(c) 314 m/s
(d) 3000 m/s

✔ **Answer: (b) 157 m/s**

Explanation: $\pi \times 1 \times 50$.

Q10. Blades of an impulse turbine are:

(a) Symmetrical
(b) Aerofoil
(c) Flat
(d) Curved asymmetric

✔ **Answer: (a) Symmetrical**

Explanation: Equal inlet and exit angles.

Q11. Largest axial thrust occurs in:

(a) Impulse
(b) Reaction
(c) De Laval
(d) Curtis

✔ **Answer: (b) Reaction**

Explanation: Pressure difference across blades.

Q12. Rateau turbine is:

(a) Velocity compounded
(b) Pressure compounded
(c) Reaction
(d) Single stage

✔ **Answer: (b) Pressure compounded**

Explanation: Rateau.

Q13. Degree of reaction of an impulse turbine:

(a) 0
(b) 0.5
(c) 1
(d) 0.25

✔ **Answer: (a) 0**

Explanation: No pressure drop in moving blades.

Q14. Speed of a De Laval turbine is about:

(a) 300 rpm
(b) 3000 rpm
(c) 30,000 rpm
(d) 300,000 rpm

✔ **Answer: (c) 30,000 rpm**

Explanation: Very high.

Q15. u = 150 m/s, ΣC_w = 400 m/s, 5 kg/s — power:

(a) 30 kW
(b) 300 kW
(c) 3000 kW
(d) 60 kW

✔ **Answer: (b) 300 kW**

Explanation: 5 × 150 × 400 W.

SSC JE / UPSSSC JE Previous Year Questions — Steam Turbines

Q1. De Laval turbine is a:

SSC JE (Previous Year)

(a) Simple impulse turbine
(b) Reaction turbine
(c) Velocity compounded
(d) Pressure compounded

✔ **Answer: (a) Simple impulse turbine**

Explanation: Single-stage impulse.

Q2. Example of a 50% reaction turbine:

SSC JE (Previous Year)

(a) Curtis
(b) Parsons
(c) De Laval
(d) Rateau

✔ **Answer: (b) Parsons**

Explanation: Parsons.

Q3. In an impulse turbine steam expands in:

SSC JE (Previous Year)

(a) Nozzles only
(b) Moving blades only
(c) Both
(d) Condenser

✔ **Answer: (a) Nozzles only**

Explanation: Nozzles.

Q4. Compounding in steam turbines reduces:

SSC JE (Previous Year)

(a) Efficiency
(b) Rotor speed
(c) Steam pressure
(d) Number of blades

✔ **Answer: (b) Rotor speed**

Explanation: Speed reduces.

Q5. Degree of reaction is defined as:

SSC JE (Previous Year)

(a) Enthalpy drop in moving blades ÷ total stage drop (b) Fixed blade drop ÷ total

(c) Work ÷ heat (d) Pressure ratio

✔ **Answer: (a) Enthalpy drop in moving blades ÷ total stage drop**

Explanation: Definition.

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